



DOC333: Introduction to Programming Principles

Comprehensive Exam Review: Questions & Detailed Solutions

A4-Formatted Study Document



PART I

True / False Fundamentals

Assessing core concepts of Python syntax, lists, and indentation.



TRUE / FALSE: QUESTIONS 10, 11, 9, 4

Question Content	Answer & Rationale
Q10: The following represents the same list: <code>['a', 'b', 'c']</code> & <code>['b', 'a', 'c']</code>	FALSE Lists are ordered. Since the index of elements differs, Python evaluates them as unequal.
Q11: A list may contain any type of object except another list.	FALSE Python supports "Nested Lists." A list is a valid object type to be stored inside another list.
Q9: A bracket is used to define a block of code in Python.	FALSE Python uses Indentation (whitespace) to define scope/blocks, not curly braces.
Q4: If <code>pop()</code> results in <code>[3,4,5,20,5,25,1]</code> , was the original list <code>[3,4,5,20,5,25,1,3]</code> ?	TRUE <code>pop()</code> removes and returns the last item. Adding '3' back to the end restores the original.

TRUE / FALSE: QUESTIONS 5, 6, 12, 13

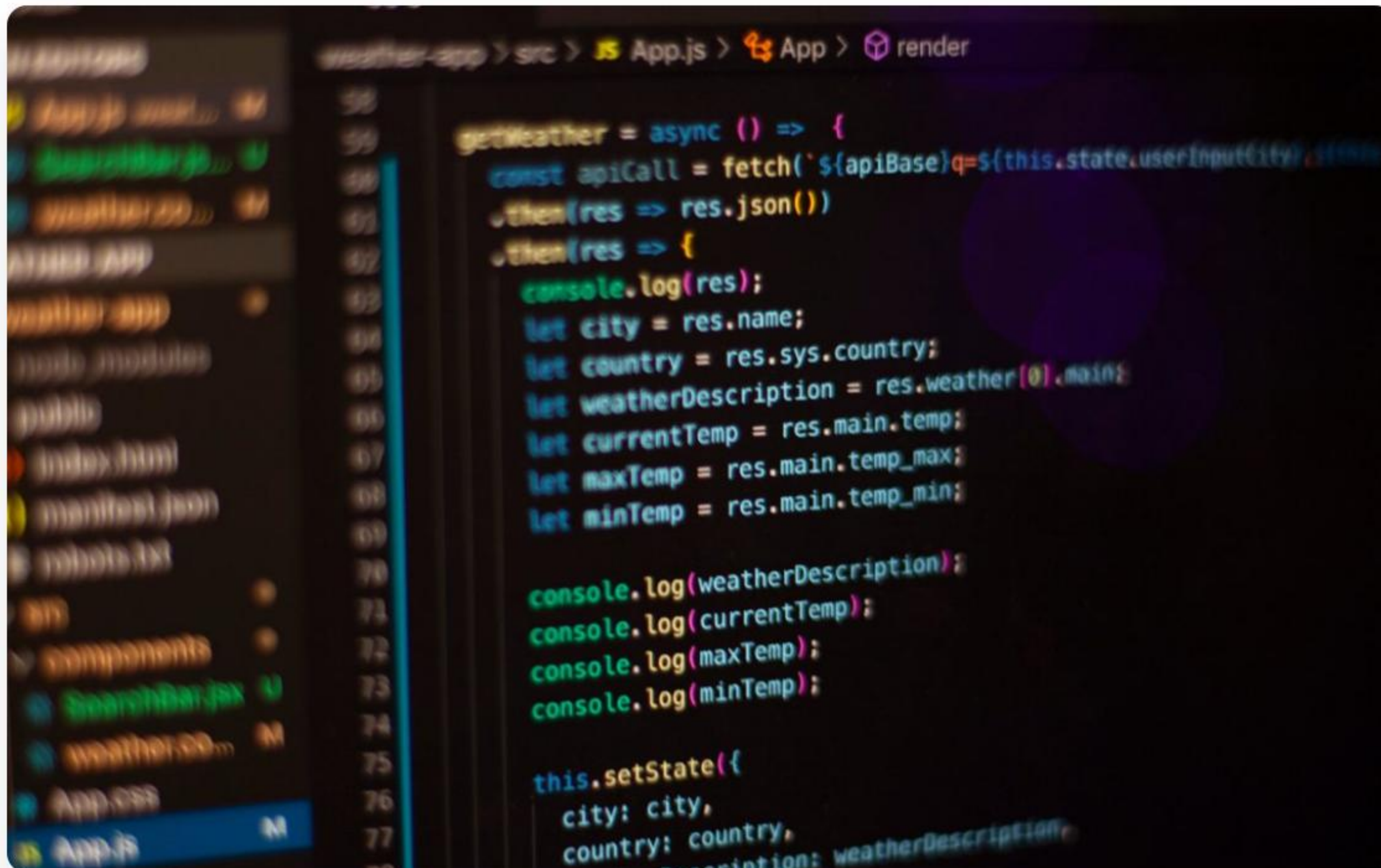
Question Content	Answer & Rationale
Q5: Loop prints "answer" each time through the loop (while $n > 0$, $n = n + 1$).	FALSE The print statement is not indented; it executes only <i>after</i> the loop (which is actually infinite).
Q6: There is no conceptual limit to the size of a list.	TRUE Theoretically, lists are dynamic. The only real limit is physical system memory (RAM).
Q12: Nested loop (4 iterations x 5 iterations) prints exactly 20 stars.	TRUE Outer loop runs 4 times; inner loop runs for each item in a 5-element list. $4 \times 5 = 20$.
Q13: Python is a case-sensitive language when dealing with identifiers.	TRUE Value and value are distinct variables in the Python interpreter.

PART II

Algorithmic Logic

Focus on List Intersections and Sequential Pseudocode.

LIST SELECTION & INTERSECTION (QUESTION 7)



```
weather-app > src > App.js > App > render
58
59 getWeather = async () => {
60   const apiCall = fetch(`${apiBase}q=${this.state.userInputCity}, units=metric`);
61   .then(res => res.json())
62   .then(res => {
63     console.log(res);
64     let city = res.name;
65     let country = res.sys.country;
66     let weatherDescription = res.weather[0].main;
67     let currentTemp = res.main.temp;
68     let maxTemp = res.main.temp_max;
69     let minTemp = res.main.temp_min;
70
71     console.log(weatherDescription);
72     console.log(currentTemp);
73     console.log(maxTemp);
74     console.log(minTemp);
75
76     this.setState({
77       city: city,
78       country: country,
79       description: weatherDescription,
```

Objective: Display 2, 12, 45, 56

The code uses a loop to find common elements (intersection) between two lists L1 and L2.

Correct Configuration:

L1: Must contain the set {2, 12, 45, 56}.

L2: Must also contain {2, 12, 45, 56}.

Ensure no other duplicates exist if exactly these four are required. Based on exam options, select the pair where these match.

ALGORITHM COMPLETION (QUESTION 14)

Pseudocode Completion

To calculate the sum and average of 10 numbers:

- 1 Step 2: COUNT = 0 (Initialize)
- 2 Step 4: INPUT NUM (Acquire data)
- 3 Step 5: SUM = SUM + NUM (Accumulate)
- 4 Step 8: GOTO Step 4 (Iterate)
- 5 Step 10: AVG = SUM / COUNT (Finalize)



FINAL SUMMARY

Key Takeaways:

Python uses **indentation** for code blocks—this is a frequent exam trap.

List Equality requires both same elements and same order.

Algorithm Logic requires careful initialization of counters and accumulators before the loop starts.

IMAGE SOURCES



<https://upload.wikimedia.org/wikipedia/commons/thumb/c/c3/Python-logo-notext.svg/1280px-Python-logo-notext.svg.png>

Source: en.wikipedia.org



<https://www.intelogy.co.uk/wp-content/uploads/2025/05/Pro-code-blog-III-.jpg>

Source: www.intelogy.co.uk



https://www.slideteam.net/media/catalog/product/cache/1280x720/d/a/data_cleaning_flowchart_using_gear_folder_server_icons_Slide01.jpg

Source: www.slideteam.net
